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Assignment 4

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Discipline: MAC0623 - 3D Interaction in Mixed Realities

Repository: https://github.com/adminLTR/A4-XR

Demo: https://adminltr.github.io/A4-XR/

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1. Task and Environment

The task is 3-DoF docking on a real horizontal surface. The phone is a monoscopic window onto the room. The user points the camera at a table, a green reticle appears when the hit-testing finds a plane, and a tap anchors a workspace there. After that, hit-testing is no longer used to move the workspace: the cube and the ghost stay on that plane.

This build uses **one cube**, 20 cm on a side. Opposite faces share a color (blue, yellow, red). The ghost is a translucent copy of the same cube, placed at the workspace origin with a random position. The movable cube starts about 18 cm to the side, also with a random position.

Docking succeeds when the distance is ≤ 2 cm and rotation is ≤ 10°. The HUD shows live position and rotation error. When both are inside tolerance the ghost glows green. The trial then ends automatically once the fingers leave the glass.

2. Manipulation Techniques

2.1. Touch gestures (baseline)

After the cube is selected, one finger maps onto the table plane. A ray is cast from the camera through the touch point; the intersection with the workspace plane becomes the position of the cube, with a grab offset so the cube does not jump under the finger.

Two fingers rotate yaw only. The twist angle between the two contact points is applied to rotation.y. Translation and rotation therefore share the same glass surface and can couple: a two-finger pose that is not a pure twist also moves the cube, and the fingers cover the ghost while the user is trying to read the colors.

Selection is a tap-to-ray pick. In this one-cube build the piece is already highlighted at trial start, so the selection step does not add a search among distractors.

2.2. Carry

After the user selects a piece, they hold an on-screen button. While the button is held, the selected object follows the intersection of the phone's forward ray with the table plane: moving the phone moves the object. Releasing the button parks it. Rotation is not a two-finger twist. It uses on-screen ↺ / ↻ buttons, so translation and rotation are separate modes.

- Isomorphic: The phone is already a tracked 6-DoF window onto the room. Using that motion to carry the object avoids the extra mapping from a 2D finger path onto the 3D plane.
- Separated DoF: two-finger gestures on a phone couple drag and twist, and the fingers cover the target (occlusion).
- Predicted trade-off: Carry should be **faster**, especially larger moves, because the user can walk the phone along the table instead of dragging a finger that

also hides the ghost. The cost is **arm fatigue** (the phone must stay aimed at the table)

3. Instrumentation & Study Design

Timing starts at the first move of the cube (first drag, twist, carry, or yaw nudge) and ends when the cube is docked. Position error is the horizontal distance in centimetres between cube and ghost. Rotation error is the smallest of the direct yaw difference and the 180°- equivalent difference, in degrees.

The file submitted with this report is `data.csv`.

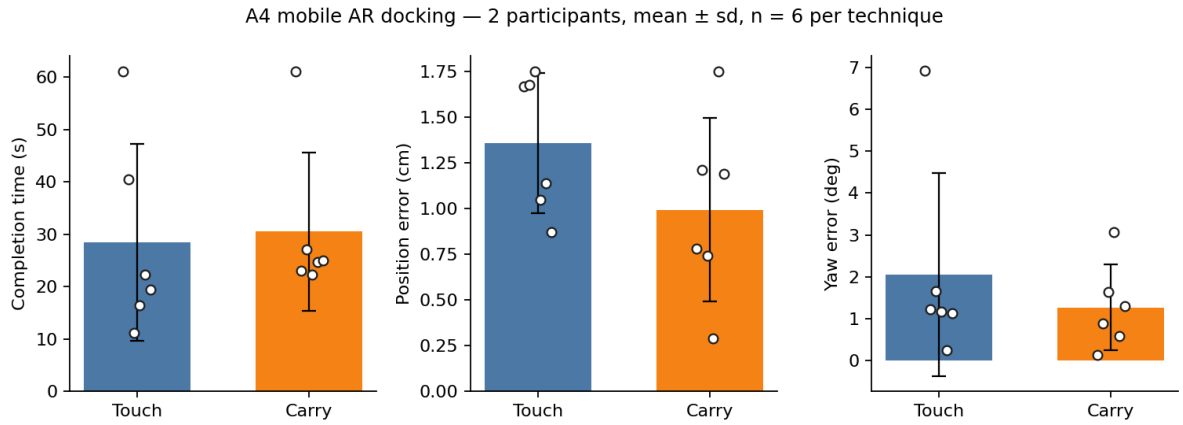
Field	Meaning
participant_id	Anonymized ID
technique	touch or carry
trial	Trial 1, 2 or 3
time_s	Completion time in seconds
dpos_cm	Final position error in cm
drot_deg	Final rotation error in degrees

4. Data collection

The assignment asks for at least 2 participants × 2 techniques × 3 trials, with technique order counterbalanced, plus a 1–5 arm-fatigue question after each technique.

That is 2 × 2 × 3 = 12 rows. The CSV lists touch then carry for both people.. Arm fatigue was asked for by the assignment but was not recorded in this CSV, so it cannot be reported as a number.

5. Results



Every logged trial finished inside tolerance. Pooled mean time is 28.4s for touch and 30.5 s for carry: carry is about 2 s slower. That gap is not shared. P001's means are almost equal (37.5 s touch, 37.0 s carry). P002 is faster with touch (19.3 s vs 23.9 s). P001's first trial of each technique is 61 s; after that touch keeps falling to 11 s while carry stays in the mid-20 s.

6. Conclusion

The prediction was that carry would be faster than touch, especially on large translations, and that the cost would be higher rotation error and higher arm fatigue.

Pooled mean time is 28.4 s for touch and 30.5 s for carry. Carry is slower, not faster. P001 is a tie (37.5 s vs 37.0 s). P002 is faster with touch (19.3 s vs 23.9 s). Carry was the second block for both people, so it had extra practice, and it still lost on time. The clutch is a plausible reason: every carry dock needs a hold, a phone move, a release, then several ↺ / ↻ steps, while a practiced one-finger drag plus a short twist can finish in one motion. P001's trial 3 shows that once the mapping is learned: 11 s touch vs 27 s carry.

The rotation-cost half of the prediction is mixed and does not save the speed claim. Pooled rotation error is lower for carry (1.27° vs 2.06°), the opposite of "buttons will be less accurate," but that is P001's first touch trial (6.91°). P002 matches the predicted cost: carry 1.65° vs touch 0.85°. Pooled position error is lower for carry (0.99 cm vs 1.36 cm), again only for P001. P002's position errors are almost the same. The assignment's primary comparison was time, and time did not move in carry's favour.

The draft predicted higher arm fatigue for carry because the phone must stay aimed at the table while translating. After each technique block, both participants answered "How tired are your arms?" on the 1–5 scale the assignment specifies. Both reported little or no arm fatigue for touch and for carry (qualitatively: "no" / not tired). That does not support the fatigue side of the predicted trade-off. What they did mention verbally was difficulty docking the cube, getting fine position and rotation within tolerance felt fiddly, especially controlling movement precisely, not soreness or tiredness from holding the phone.

We have to remember one thing: two participants, three trials, both people did touch first, a one-cube task with no distractor selection, and a reduced CSV. These numbers describe these two sessions. They do not estimate a population difference.